specguard report

`specguard/examples/arp4754b_appendix_e.py`

May 10, 2026

1 Encoded collections

`PSSA_REQUIREMENTS`

`E12-1` Airplane electrical power 1 independent from airplane electrical power 2

$$\neg(\text{lost\_elec\_input\_1} \land \text{lost\_elec\_input\_2})$$

`E12-2` NORMAL Mode braking function independent from ALTERNATE Mode braking function, for total loss failure condition

$$\neg(\text{failed\_normal\_braking} \land \text{failed\_alternate\_braking})$$

`E12-3` The NMVs and their control are independent from the SOV and its control such that no single failure results in uncommanded braking

$$(\text{single\_failure} \Rightarrow \neg\text{inadvertent\_braking\_takeoff})$$

`E13-1` The probability of BSCU failure resulting in loss of a valid braking command output to the NMV shall not exceed 2.0E-04 per flight

$$\text{p\_bscu\_loss\_braking\_cmd} \leq 2 \times 10^{-4}$$

`E13-2` The probability of BSCU failure resulting in unannunciated erroneous braking command to the NMV shall not exceed 2.0E-04 per flight

$$\text{p\_bscu\_erroneous\_cmd} \leq 2 \times 10^{-4}$$

`E13-3` The probability of BSCU failure resulting in the loss of command to open the SOV shall not exceed 2.0E-04 per flight

$$\text{p\_bscu\_loss\_sov\_cmd} \leq 2 \times 10^{-4}$$

`E13-4` The probability of BSCU failure resulting in unintended closure of the S/ASV shall not exceed 2.0E-04 per flight

$$\text{p\_bscu\_unintended\_sasv} \leq 2 \times 10^{-4}$$

`E13-5` The SOV and NMV commands shall be provided by the BSCU upon loss of either airplane electrical power input

$$(((\text{lost\_elec\_input\_1} \wedge \neg \text{lost\_elec\_input\_2}) \Rightarrow (\text{bscu\_provides\_sov\_cmd} \wedge \text{bscu\_provides\_nmv\_cmd})) \wedge ((\text{lost\_elec\_input\_2} \wedge \neg \text{lost\_elec\_input\_1}) \Rightarrow (\text{bscu\_provides\_sov\_cmd} \wedge \text{bscu\_provides\_nmv\_cmd})))$$

`E13-6` When "HYD 1 Enable" output is enabled, then "Alt/Emer Ctrl" output shall be disabled

$$(\text{hyd1\_enable\_on} \Rightarrow \neg \text{alt\_emer\_ctrl\_on})$$

`E13-7` When "HYD 1 Enable" output is disabled, then "Alt/Emer Ctrl" output shall be enabled

$$(\neg \text{hyd1\_enable\_on} \Rightarrow \text{alt\_emer\_ctrl\_on})$$

`E13-8` No single failure shall cause erroneous NMV command and inhibit the SOV function

$$\neg(\text{erroneous\_nmv\_cmd} \wedge \text{sov\_function\_inhibited})$$

`E13-9` The wheel brake command function of the BSCU shall be developed to FDAL A

$$\text{bscu\_cmd\_fdal\_a}$$

`E14-1` The probability of 'Loss of Normal Braking System Hydraulic Equipment' will be less than 3.3E-05 per flight

$$\text{p\_loss\_normal\_hyd\_equip} < 3.3 \times 10^{-5}$$

`E14-2` The probability of 'Loss of Alternate Braking System Hydraulic Equipment' will be less than 3.3E-05 per flight

$$\text{p\_loss\_alt\_hyd\_equip} < 3.3 \times 10^{-5}$$

`E14-3` The probability of 7 or more wheel speed sensors erroneous or inoperative will be less than 1.0E-07 per flight

$$\text{p\_seven\_or\_more\_sensors\_per\_flight} < 1 \times 10^{-7}$$

`E14-4` The probability of loss of an airplane electrical power bus will be less than 1.0E-04 per flight

$$\text{p\_loss\_elec\_bus\_per\_flight} < 1 \times 10^{-4}$$

`E14-5` The probability of loss of a Left brake pedal position input will be less than 1.0E-04 per flight

$$\text{p\_loss\_pedal\_per\_flight} < 1 \times 10^{-4}$$

`E14-6` Airplane electrical power bus 1 is independent from airplane electrical power bus 2

$$\neg(\text{lost\_elec\_input\_1} \wedge \text{lost\_elec\_input\_2})$$

`E14-7` HYD 1 hydraulic system is independent from HYD 2 hydraulic system

$$\neg(\text{failed\_hyd\_1} \wedge \text{failed\_hyd\_2})$$

ALT_PSSA_REQUIREMENTS

E28-WBS-ASMP-1 The probability of 'Loss of Normal Braking System Hydraulic Equipment' will be less than 3.3E-05 per flight

$$\text{p\_loss\_normal\_hyd\_equip} < 3.3 \times 10^{-5}$$

E28-WBS-ASMP-2 The probability of 'Loss of Alternate Braking System Hydraulic Equipment' will be less than 3.3E-05 per flight

$$\text{p\_loss\_alt\_hyd\_equip} < 3.3 \times 10^{-5}$$

E28-WBS-ASMP-3 The probability of 7 or more wheel speed sensors erroneous or inoperative will be less than 1.0E-07 per flight

$$\text{p\_seven\_or\_more\_sensors\_per\_flight} < 1 \times 10^{-7}$$

E28-WBS-ASMP-4 The failure rate for loss of an airplane electrical power bus will be less than 1.0E-04 per hour of flight

$$\text{lambda\_loss\_elec\_bus\_per\_hour} < 1 \times 10^{-4}$$

E28-WBS-ASMP-5 The failure rate for loss of a left brake pedal position input will be less than 1.0E-04 per hour of flight

$$\text{lambda\_loss\_pedal\_per\_hour} < 1 \times 10^{-4}$$

E28-WBS-ASMP-6 Airplane electrical power bus 1 is independent from airplane electrical power bus 2

$$\neg(\text{lost\_elec\_input\_1} \wedge \text{lost\_elec\_input\_2})$$

E28-WBS-ASMP-7 HYD 1 hydraulic system is independent from HYD 2 hydraulic system

$$\neg(\text{failed\_hyd\_1} \wedge \text{failed\_hyd\_2})$$

SPEC_REQUIREMENTS

S18-WBS-R-0020 The Wheel Brake System shall have a means to decelerate the wheels on the ground

$$\text{wbs\_has\_decelerate\_means}$$

S18-WBS-R-0021 The Wheel Brake System shall be capable of decelerating the S18 airplane to a complete stop/to taxi speed in 2000 feet when wheel brakes, high lift speed brakes and reverse thrust are available, including when at maximum landing weight

$$\text{stopping\_distance\_ft} \leq 2000$$

S18-WBS-R-0041 The Wheel Brake System shall provide directional control on ground by a differential braking function

$$\text{wbs\_differential\_braking}$$

`S18-WBS-R-0042` The Wheel Brake System shall provide a parking brake to prevent airplane motion on the ground

wbs_parking_brake

`S18-WBS-R-0043` The Wheel Brake System shall provide autobraking capability during landing and RTO

wbs_autobraking

`S18-WBS-R-0044` The Wheel Brake System shall provide anti-skid braking

wbs_antiskid

`S18-WBS-R-0045` The Wheel Brake System shall provide hydraulic brake control

wbs_hyd_brake_control

`S18-WBS-R-0046` The Wheel Brake System shall override the autobrake when the commanded by the flight crew

wbs_overrides_autobrake_on_crew

`S18-WBS-R-0047` The Wheel Brake System shall meet safety requirements while operating in an average atmospheric radiation environment per the IEC 62396 standard with an altitude of 40000 feet and a latitude of 45 degrees

wbs_radiation_qualified

`S18-WBS-R-0049` The Wheel Brake System shall be controlled and monitored by a computer system called Brake System Control Unit

wbs_controlled_by_bscu

`S18-WBS-R-0050` Each wheel brake shall have separate hydraulic power supply lines

each_wheel_separate_hyd_lines

`S18-WBS-R-0052` Each hydraulic Wheel Brake System circuit shall have meter valves, anti-skid valves and hydraulic fuses

each_circuit_valves_fuses

`S18-WBS-R-0055` Anti-skid system shall be capable to prevent skidding of the tires by reducing the pressure applied to the brakes

antiskid_prevents_skidding

`S18-WBS-R-0062` The Wheel Brake System shall include an emergency accumulator to supply hydraulic power to the wheel brakes

has_emergency_accumulator

`S18-WBS-R-0065` The emergency accumulator shall provide a minimum of 1800 psi

$$\text{accumulator\_pressure\_psi} \geq 1800$$

`S18-WBS-R-0066` The Alternate/Emergency Brake System hydraulic equipment and piping shall be installed aft of the engine 1 UERF trajectory envelope

$$\text{alt\_emer\_aft\_of\_uerf}$$

`S18-WBS-R-0100` The Wheel Brake System decelerate the wheels on the ground function shall be developed as FDAL A

$$\text{wbs\_decelerate\_fdal\_a}$$

`S18-WBS-R-0120` No single failure or event shall cause the complete loss of hydraulic power for both the WBS Normal and Alternate/Emergency Brake Systems

$$\neg(\text{failed\_normal\_braking} \land \text{failed\_alternate\_braking})$$

`S18-WBS-R-0130` No single failure or event shall cause the complete loss of electrical power for both the WBS Normal and Alternate/Emergency Brake Systems

$$\neg(\text{lost\_elec\_input\_1} \land \text{lost\_elec\_input\_2})$$

`S18-WBS-R-0150` Complete loss of decelerate the wheels on the ground function shall be less probable than 1.0E-07 for a landing

$$\text{p\_total\_loss\_decelerate} < 1 \times 10^{-7}$$

`S18-WBS-R-0200` Two redundant control lanes shall be provided between the EBU and each of the two Alternate/Emergency Meter Valves

$$\text{ebu\_two\_redundant\_lanes}$$

`S18-WBS-R-0326` No single failure shall result in inadvertent wheel braking of all wheels during takeoff roll

$$(\text{single\_failure} \Rightarrow \neg\text{inadvertent\_braking\_takeoff})$$

`S18-WBS-R-0508` The Wheel Brake System shall have at least two independent hydraulic pressure sources

$$\text{wbs\_two\_independent\_hyd\_sources}$$

`S18-WBS-R-0509` The Wheel Brake System shall have dual BSCU command functions

$$\text{wbs\_dual\_bscu\_cmd}$$

`S18-WBS-R-0510` The rudder and nose wheel steering functions shall not be implemented by the Wheel Brake System

$$\text{wbs\_no\_rudder\_or\_nws}$$

`S18-WBS-R-0631` The Wheel Brake System shall indicate individual brake temperature

$$\text{wbs\_indicates\_brake\_temp}$$

`S18-WBS-R-0632` The Wheel Brake System shall control individual brake pressure

$$\text{wbs\_individual\_brake\_pressure}$$

`S18-WBS-ICD-1002` The brake actuators shall respond to a BSCU command within 5 ms of receiving it

$$\text{actuator\_response\_time\_ms} \leq 5$$

`S18-WBS-R-1613` Hydraulic pressure shall be controlled for weight, autobrake mode, ground speed, wheel rotation, brake temperature and deceleration rate in accordance with graphs given in S18 brake force analysis AAS18-XXX

$$\text{hyd\_pressure\_per\_analysis}$$

`S18-WBS-R-2971` Wheel Brake System shall have one NORMAL operating mode

$$\text{wbs\_one\_normal\_mode}$$

`S18-WBS-R-2972` Wheel Brake System shall have one ALTERNATE operating mode

$$\text{wbs\_one\_alternate\_mode}$$

`S18-WBS-R-2973` Operation of the Alternate/Emergency Brake System shall be precluded when the Normal Brake System is in use

$$(\text{normal\_mode\_active} \Rightarrow \neg\text{alternate\_mode\_active})$$

`S18-WBS-R-2974` An emergency accumulator shall provide hydraulic power for EMERGENCY Wheel Brake System operating mode

$$\text{accumulator\_powers\_emergency}$$

`S18-WBS-R-2975` The emergency accumulator shall be attached to the HYD 2 hydraulic line between the Selector Valve and the Alternate/Emergency Meter Valve

$$\text{accumulator\_attached\_to\_hyd2}$$

`S18-WBS-R-2976` Wheel Brake System shall charge the emergency accumulator prior to normal operation

$$\text{wbs\_charges\_accumulator\_pre\_normal}$$

`S18-WBS-R-2986` The wheel brake command function of the BSCU shall be developed as FDAL A

$$\text{bscu\_cmd\_fdal\_a}$$

`S18-WBS-R-2997` The BSCU shall have two independent command channels

$$\text{bscu\_two\_independent\_channels}$$

`S18-WBS-R-2999` The BSCU shall have two independent electrical power sources

$$\text{bscu\_two\_independent\_power}$$

`S18-WBS-R-3011` Brake pedals shall be independent from the rudder control

$$\text{brake\_pedals\_indep\_rudder}$$

`S18-WBS-R-3213` Normal Brake System shall be powered from the airplane's HYD 1 hydraulic system

$$(\text{failed\_hyd\_1} \Rightarrow \text{failed\_normal\_braking})$$

`S18-WBS-R-3224` Alternate/Emergency Brake System shall be powered from the airplane's HYD 2 hydraulic system

$$(\text{failed\_hyd\_2} \Rightarrow \text{failed\_alternate\_braking})$$

`S18-WBS-R-3245` ALTERNATE Mode shall be automatically selected when the NORMAL Mode fails

$$(\text{normal\_mode\_failed} \Rightarrow \text{alternate\_mode\_active})$$

`S18-WBS-R-6104` The probability of BSCU failure resulting in loss of a valid braking command output to the NMV shall not exceed 2.0E-04 per flight

$$\text{p\_bscu\_loss\_braking\_cmd} \leq 2 \times 10^{-4}$$

`S18-WBS-R-6105` The probability of BSCU failure resulting in unannunciated erroneous braking command to the NMV shall not exceed 2.0E-04 per flight

$$\text{p\_bscu\_erroneous\_cmd} \leq 2 \times 10^{-4}$$

`S18-WBS-R-6106` The probability of BSCU failure resulting in the loss of command to open the SOV shall not exceed 2.0E-04 per flight

$$\text{p\_bscu\_loss\_sov\_cmd} \leq 2 \times 10^{-4}$$

`S18-WBS-R-6107` The probability of BSCU failure resulting in unintended closure of the S/ASV shall not exceed 2.0E-04 per flight

$$\text{p\_bscu\_unintended\_sasv} \leq 2 \times 10^{-4}$$

`S18-WBS-R-6108` The SOV and NMV commands shall be provided by the BSCU upon loss of either airplane electrical power input

$$\begin{aligned}&(((\text{lost\_elec\_input\_1} \wedge \neg \text{lost\_elec\_input\_2}) \Rightarrow (\text{bscu\_provides\_sov\_cmd} \wedge \text{bscu\_provides\_nmv\_cmd}))\\ &\wedge ((\text{lost\_elec\_input\_2} \wedge \neg \text{lost\_elec\_input\_1}) \Rightarrow (\text{bscu\_provides\_sov\_cmd} \wedge \text{bscu\_provides\_nmv\_cmd})))\end{aligned}$$

`S18-WBS-R-6109` When "HYD 1 Enable" output is enabled, then "Alt/Emer Ctrl" output shall be disabled

$$(\text{hyd1\_enable\_on} \Rightarrow \neg\text{alt\_emer\_ctrl\_on})$$

`S18-WBS-R-6110` No single failure shall cause erroneous NMV command and inhibit the SOV function

$$\neg(\text{erroneous\_nmv\_cmd} \wedge \text{sov\_function\_inhibited})$$

`S18-WBS-R-6111` The probability of 'Loss of Normal Brake System Hydraulic Components' shall be less than 3.3E-05 per flight

$$\text{p\_loss\_normal\_hyd\_equip} < 3.3 \times 10^{-5}$$

`S18-WBS-R-6112` The probability of 'Loss of Alternate/Emergency Brake System Hydraulic Components' shall be less than 3.3E-05 per flight

$$\text{p\_loss\_alt\_hyd\_equip} < 3.3 \times 10^{-5}$$

`BSCU_PSSA_REQUIREMENTS`

`BSCU-001` Channel #1 shall have physical independence from Channel #2

$$\text{channel1\_phys\_indep\_channel2}$$

`BSCU-002` Within each channel, the command function shall have physical independence from the monitor function

$$\text{cmd\_phys\_indep\_monitor}$$

`BSCU-003` The COMX software and MONX software shall be developed using different software requirement specifications

$$\text{comx\_monx\_diff\_specs}$$

`BSCU-004` Within each channel, a power supply monitor shall shutoff the power supply when any voltage is detected to be out of specification

$$(\text{voltage\_out\_of\_spec} \Rightarrow \text{power\_supply\_shutoff})$$

`BSCU-005` Within each channel, the operational state of the power supply shall have no effect on the operational state of the power supply monitor

$$\text{power\_supply\_state\_indep\_monitor}$$

`BSCU-006` Upon Channel #1 indicating an 'invalid' state, Channel #2 shall command the 'Normal Control' output

$$(\text{channel1\_invalid} \Rightarrow \text{channel2\_cmd\_normal\_ctrl})$$

`BSCU-007` Upon both Channel #1 and Channel #2 indicating 'invalid' states, the 'HYD 1 Enable' output shall be disabled

$$((\text{channel1\_invalid} \wedge \text{channel2\_invalid}) \Rightarrow \neg\text{hyd1\_enable\_on})$$

BSCU-008 Upon either Channel #1 or Channel #2 indicating a validity error, a WBS annunciation shall be output from the BSCU

$$((\text{channel1\_invalid} \vee \text{channel2\_invalid}) \Rightarrow \text{wbs\_annunciation\_out})$$

BSCU-009 At power-up the BSCU shall perform a self-test of Channel #2 including the ability to switch control from Channel #1 to Channel #2

$$\text{power\_up\_self\_tests\_ch2}$$

BSCU-010 At power-up each channel of the BSCU shall implement a test of its power supply monitor

$$\text{power\_up\_tests\_psm}$$

BSCU-011 If the power-up test fails, a WBS annunciation shall be output from the BSCU

$$(\text{power\_up\_test\_failed} \Rightarrow \text{wbs\_annunciation\_out})$$

BSCU_ASSUMPTIONS_TO_WBS

E23-1 When 'HYD 1 Enable' is disabled, the BSCU brake control outputs are not used

$$(\neg\text{hyd1\_enable\_on} \Rightarrow \neg\text{bscu\_brake\_outputs\_in\_use})$$

E23-2 A BSCU maintenance action will be initiated, if the WBS Status indicates a failure

$$(\text{wbs\_status\_failure} \Rightarrow \text{bscu\_maintenance\_initiated})$$

BSCU_SPEC_REQUIREMENTS

S18-BSCU-R-0001 The wheel brake command function of the BSCU shall be developed as FDAL A

$$\text{bscu\_cmd\_fdal\_a}$$

S18-BSCU-R-0002 The probability of BSCU failure resulting in loss of a valid braking command output to the NMV shall not exceed 2.0E-04 per flight

$$\text{p\_bscu\_loss\_braking\_cmd} \leq 2 \times 10^{-4}$$

S18-BSCU-R-0003 The probability of BSCU failure resulting in unannunciated erroneous braking command to the NMV shall not exceed 2.0E-04 per flight

$$\text{p\_bscu\_erroneous\_cmd} \leq 2 \times 10^{-4}$$

S18-BSCU-R-0004 The probability of BSCU failure resulting in the loss of command to open the SOV shall not exceed 2.0E-04 per flight

$$\text{p\_bscu\_loss\_sov\_cmd} \leq 2 \times 10^{-4}$$

`S18-BSCU-R-0005` The probability of BSCU failure resulting in unintended closure of the S/ASV shall not exceed 2.0E-04 per flight

$$\text{p\_bscu\_unintended\_sasv} \leq 2 \times 10^{-4}$$

`S18-BSCU-R-0006` The SOV and NMV commands shall be provided by the BSCU upon loss of either airplane electrical power input

$$(((\text{lost\_elec\_input\_1} \wedge \neg \text{lost\_elec\_input\_2}) \Rightarrow (\text{bscu\_provides\_sov\_cmd} \wedge \text{bscu\_provides\_nmv\_cmd})) \wedge ((\text{lost\_elec\_input\_2} \wedge \neg \text{lost\_elec\_input\_1}) \Rightarrow (\text{bscu\_provides\_sov\_cmd} \wedge \text{bscu\_provides\_nmv\_cmd})))$$

`S18-BSCU-R-0007` When "HYD 1 Enable" output is enabled, then "Alt/Emer Ctrl" output shall be disabled

$$(\text{hyd1\_enable\_on} \Rightarrow \neg \text{alt\_emer\_ctrl\_on})$$

`S18-BSCU-R-0008` No single failure shall cause erroneous NMV command and inhibit the SOV function

$$\neg(\text{erroneous\_nmv\_cmd} \wedge \text{sov\_function\_inhibited})$$

`S18-BSCU-R-0009` When 'HYD 1 Enable' is disabled, the BSCU brake control outputs shall be disabled

$$(\neg \text{hyd1\_enable\_on} \Rightarrow \neg \text{bscu\_brake\_outputs\_in\_use})$$

`S18-BSCU-R-0010` BSCU shall calculate the required hydraulic pressure as a function of weight, autobrake mode, ground speed, wheel rotation, brake temperature and deceleration rate in accordance with graphs given in S18 brake force analysis AAS18-XXX

$$\text{hyd\_pressure\_per\_analysis}$$

`S18-BSCU-R-0011` BSCU shall control the meter valves

$$\text{bscu\_controls\_meter\_valves}$$

`S18-BSCU-R-0012` BSCU shall control the anti-skid valves

$$\text{bscu\_controls\_antiskid\_valves}$$

`S18-BSCU-R-0013` BSCU shall control the shut off valves

$$\text{bscu\_controls\_shutoff\_valves}$$

`S18-BSCU-R-0020` BSCU shall control the valves to charge the emergency accumulator prior to normal operation

$$\text{bscu\_charges\_accumulator\_pre\_normal}$$

`S18-BSCU-R-0027` Each BSCU shall have two command channels

$$\text{bscu\_each\_two\_command\_channels}$$

`S18-BSCU-R-0051` Each BSCU channel shall have a monitoring unit

bscu_each_channel_has_monitoring

`S18-BSCU-R-0075` Each BSCU channel shall have a control unit

bscu_each_channel_has_control

`S18-BSCU-R-0099` Each BSCU channel shall have independent power

bscu_each_channel_indep_power

`S18-BSCU-R-0123` Each BSCU channel shall have independent pedal input to both control and monitoring units

bscu_each_channel_indep_pedal

`S18-BSCU-R-0124` The command channels of each BSCU shall be independent of each other

bscu_command_channels_indep

`S18-BSCU-R-0126` The BSCU hardware shall be developed to IDAL A

bscu_hw_idal_a

`S18-BSCU-R-0127` The BSCU software shall be developed to IDAL B

bscu_sw_idal_b

`S18-BSCU-R-0128` BSCU shall meet safety requirements while operating in an average atmospheric radiation environment per the IEC 62396 standard with an altitude of 40000 feet and a latitude of 45 degrees

bscu_radiation_qualified

`S18-BSCU-R-0201` Channel #1 shall have physical independence from Channel #2

channel1_phys_indep_channel2

`S18-BSCU-R-0202` Within each channel, the command function shall have physical independence from the monitor function

cmd_phys_indep_monitor

`S18-BSCU-R-0203` The COMX software and MONX software shall be developed using different software requirement specifications

comx_monx_diff_specs

`S18-BSCU-R-0204` Within each channel, a power supply monitor shall shutoff the power supply when any voltage is detected to be out of specification

(voltage_out_of_spec ⇒ power_supply_shutoff)

`S18-BSCU-R-0205` Within each channel, the operational state of the power supply shall have no effect on the operational state of the power supply monitor

$$\text{power\_supply\_state\_indep\_monitor}$$

`S18-BSCU-R-0206` Upon Channel #1 indicating an 'invalid' state, Channel #2 shall command the 'Normal Control' output

$$(\text{channel1\_invalid} \Rightarrow \text{channel2\_cmd\_normal\_ctrl})$$

`S18-BSCU-R-0207` Upon both Channel #1 and Channel #2 indicating 'invalid' states, the 'HYD 1 Enable' output shall be disabled

$$((\text{channel1\_invalid} \wedge \text{channel2\_invalid}) \Rightarrow \neg\text{hyd1\_enable\_on})$$

`S18-BSCU-R-0208` Upon either Channel #1 or Channel #2 indicating a validity error, a WBS annunciation shall be output from the BSCU

$$((\text{channel1\_invalid} \vee \text{channel2\_invalid}) \Rightarrow \text{wbs\_annunciation\_out})$$

`S18-BSCU-R-0209` At power-up, the BSCU shall perform a self-test of Channel #2 including the ability to switch control from Channel #1 to Channel #2

$$\text{power\_up\_self\_tests\_ch2}$$

`S18-BSCU-R-0210` At power-up, each channel of the BSCU shall implement a test of its power supply monitor

$$\text{power\_up\_tests\_psm}$$

`S18-BSCU-R-0211` If the power-up test fails, a WBS annunciation shall be output from the BSCU

$$(\text{power\_up\_test\_failed} \Rightarrow \text{wbs\_annunciation\_out})$$

AIRPLANE_REQUIREMENTS

`S18-ACFT-R-1000` The S18 airplane shall have a means to decelerate on ground

$$\text{airplane\_decelerate\_on\_ground}$$

`S18-ACFT-R-1100` The S18 airplane shall have a means to decelerate the wheels on ground

$$\text{airplane\_decelerate\_wheels\_on\_ground}$$

`S18-ACFT-R-1110` The S18 airplane shall have pilot-controlled wheel braking capability

$$\text{airplane\_pilot\_controlled\_braking}$$

`S18-ACFT-R-1120` The S18 airplane shall have an autobraking capability during landing and RTO

$$\text{airplane\_autobraking}$$

`S18-ACFT-R-1130` The S18 airplane shall have anti-skid braking

$$\text{airplane\_antiskid}$$

`S18-ACFT-R-1140` The S18 airplane shall provide interface for Wheel Brake System status and annunciations

$$\text{airplane\_wbs\_status\_interface}$$

`S18-ACFT-R-0181` The S18 airplane shall decelerate the wheels on gear retraction

$$\text{airplane\_decelerate\_wheels\_on\_gear\_retract}$$

`S18-ACFT-R-0182` The S18 airplane shall be capable of decelerating the wheels differentially

$$\text{airplane\_decelerate\_differentially}$$

`S18-ACFT-R-0183` The S18 airplane shall have a parking brake to prevent airplane motion when parked

$$\text{airplane\_parking\_brake}$$

`S18-ACFT-R-0184` The S18 airplane shall have hydraulically driven braking

$$\text{airplane\_hydraulic\_braking}$$

`S18-ACFT-R-0185` The S18 airplane shall have an autobrake override that can be initiated by the flight crew

$$\text{airplane\_autobrake\_override}$$

`S18-ACFT-R-0186` The S18 airplane shall meet safety requirements while operating in an average atmospheric radiation environment per the IEC 62396 standard with an altitude of 40000 feet and a latitude of 45 degrees

$$\text{airplane\_radiation\_qualified}$$

`S18-ACFT-R-0933` The Wheel Brake System decelerate the wheels on the ground function shall be developed as FDAL A

$$\text{airplane\_decelerate\_function\_fdal\_a}$$

`S18-ACFT-R-1322` No single failure or event shall result in the complete loss of wheel brake and the loss of one thrust reverser

$$\neg(\text{complete\_loss\_wheel\_brake} \wedge \text{loss\_one\_thrust\_reverser})$$

`S18-ACFT-R-1385` Complete loss of wheel braking shall be less than 1.0E-07 for a landing

$$\text{p\_complete\_loss\_wheel\_braking\_per\_landing} < 1 \times 10^{-7}$$

S18-ACFT-R-1550 Loss of power from both hydraulic subsystems powered by the engines shall not lead to complete loss of wheel braking

$$(\text{both\_engine\_hyd\_failed} \Rightarrow \neg\text{complete\_loss\_wheel\_brake})$$

S18-ACFT-R-1551 Wheel Brake System shall include an emergency accumulator to supply hydraulic power to the wheel brakes

$$\text{airplane\_has\_emergency\_accumulator}$$

S18-ACFT-R-1552 The Alternate/Emergency Brake System hydraulic equipment and piping shall be installed aft of the engine 1 UERF trajectory envelope

$$\text{alt\_emer\_aft\_of\_uerf}$$

S18-ACFT-R-1600 Two redundant control lanes shall be provided between the Electric Brake Unit (EBU) and each of the two Alternate/Emergency Meter Valves

$$\text{ebu\_two\_redundant\_lanes}$$

PASA_SAFETY_REQUIREMENTS

PASA-SR-01 Decelerate wheels function shall be developed FDAL A

$$\text{airplane\_decelerate\_function\_fdal\_a}$$

PASA-SR-05 [FF1.1] Complete loss of wheel brake shall be less than 1.0E-07 for a landing.

$$\text{p\_complete\_loss\_wheel\_braking\_per\_landing} < 1 \times 10^{-7}$$

PASA-SR-10 No single failure or event shall result in the complete loss of wheel brake and the loss of one thrust reverser.

$$\neg(\text{complete\_loss\_wheel\_brake} \wedge \text{loss\_one\_thrust\_reverser})$$

PASA-SR-12 Loss of power from both hydraulic subsystems powered by the engines shall not lead to complete loss of wheel braking.

$$(\text{both\_engine\_hyd\_failed} \Rightarrow \neg\text{complete\_loss\_wheel\_brake})$$

PASA-SR-13 The Alternate/Emergency Brake System hydraulic equipment and piping shall be installed aft of the engine 1 UERF trajectory envelope.

$$\text{alt\_emer\_aft\_of\_uerf}$$

PASA-SR-14 Two redundant control lanes shall be provided between the Electric Brake Unit and each of the two Alternate/Emergency Meter Valves.

$$\text{ebu\_two\_redundant\_lanes}$$

AIRPLANE_ASSUMPTIONS

ASMP 3.2.2-1 Overrunning the runway length above "XYZ" knots is considered a high speed overrun.

$$(\text{groundspeed\_overrun\_kt} > \text{xyz\_overrun\_threshold\_kt} \Rightarrow \text{high\_speed\_overrun})$$

ASMP 3.2.2-2 The directional aspect of asymmetric failures of deceleration systems are functionally addressed by failure conditions of the "Control direction on ground" airplane function (see function 2.3 AFHA).

$$\text{afha\_dir\_addressed\_by\_ctrl\_dir}$$

ASMP 3.2.2-3 The flight crew will divert to a suitable airfield if aware of a condition that renders the airplane incapable of landing at the originally intended destination.

$$\text{crew\_diverts\_when\_aware\_unlandable}$$

ASMP 3.2.2-4 Crew awareness is not a factor for the identified malfunction, as these are immediately evident due to aircraft behavior and do not have their effects intensified or mitigated by crew awareness features.

$$\text{malfunction\_immediately\_evident}$$

ASMP 3.2.2-5 The flight crew will not initiate a Rejected Takeoff (RTO) in response to an annunciated failure of deceleration features during takeoff due to alert suppression.

$$(\text{annunciated\_failure\_decel\_takeoff} \Rightarrow \neg\text{rto\_initiated})$$

ASMP 3.2.2-6 Taxi is performed at groundspeeds below 30 knots.

$$\text{groundspeed\_taxi\_kt} < 30$$

ASMP 3.2.2-7 Landings with failure condition in combination with environmental factors have been assessed.

$$\text{landings\_with\_env\_factors\_assessed}$$

ASMP 3.2.2-8 Failures of deceleration capability will be detected and annunciated by on-board systems.

$$(\text{failure\_decel\_capability} \Rightarrow \text{detected\_and\_annunciated\_onboard})$$

PASA-ASMP-01 It is assumed that the high speed overrun is above 30 knots and low speed overrun is below (or equal) 30 knots. This assumption has been derived from the AFHA assumption ASMP 3.2.2-1 and ASMP 3.2.2-6 for the establishment of the criteria and terms of "high-speed overrun" and "low speed overrun."

$$\begin{gathered}(\text{high\_speed\_overrun} = \text{groundspeed\_overrun\_kt} > 30 \\ \wedge\ \text{low\_speed\_overrun} = \text{groundspeed\_overrun\_kt} \leq 30)\end{gathered}$$

PASA-ASMP-02 The degraded state of systems considered in the CoFFE analysis has been defined such as half functional capability.

$$\text{degraded\_state\_half\_capability}$$

PASA-ASMP-03 Wheel Brake System uses ground detection information together with the wheel speed information. These functions are developed independently so that even if Erroneous Ground Detection Information (AGS.MF) is provided on ground, i.e., with false in-flight status, wheel braking function is available if there is correct wheel speed information.

$$((\text{erroneous\_ground\_detection} \\ \wedge \text{correct\_wheel\_speed\_info}) \Rightarrow \text{wheel\_braking\_function\_available})$$

PRA-UERF-ASSUMPTION-01 The BSCU is installed in the avionics compartment located in the nose fuselage section of the airplane, forward of the UERF area.

$$\text{bscu\_in\_avionics\_compartment\_fwd}$$

PRA-UERF-ASSUMPTION-02 The electrical lines transmitting the brake control signals from the Electric Brake Unit to the BSCU are routed "to the shortest" from one end to the other in the nose fuselage section of the airplane, thus are kept forward of the UERF area.

$$\text{ebu\_to\_bscu\_lines\_routed\_short}$$

PRA-UERF-ASSUMPTION-03 The hydraulic reservoirs and the high pressure manifolds of the hydraulic subsystems 1 and 2 are located in the fuselage, aft of the UERF area.

$$\text{hyd\_reservoirs\_aft\_of\_uerf}$$

PRA-UERF-ASSUMPTION-04 The hydraulic power distribution lines from the engine driven pumps to the respective hydraulic reservoirs and high pressure manifolds exit the engine pylons aft of the wing rear spars, then run along, aft of, the wing rear spars to the fuselage, and are kept aft of the UERF area inside the fuselage.

$$\text{hyd\_distribution\_aft\_of\_uerf}$$

PRA-UERF-ASSUMPTION-05 The hydraulic distribution lines routed to the forward part of the fuselage to supply hydraulic equipment located forward of the UERF area are fitted with appropriate isolation means located aft of the UERF area.

$$\text{hyd\_forward\_lines\_isolated}$$

PRA-UERF-ASSUMPTION-06 The power supply lines from the electrical power center(s) to the EBU are routed "to the shortest" from one end to the other in the nose fuselage section of the airplane.

$$\text{power\_supply\_lines\_routed\_short}$$

PRA-UERF-ASSUMPTION-07 In case of cabin depressurization the airplane altitude will be limited by procedure to 10000 feet.

$$\text{cabin\_depress\_alt\_limited\_10000ft}$$

WBS_SFHA_ASSUMPTIONS

SASP 1.1-1 Loss of 80% or more of deceleration capability is considered a total loss of the deceleration means.

$$(\text{deceleration\_capability\_loss\_pct} \geq 80 \Rightarrow \text{total\_loss\_deceleration})$$

SASP 1.1-2 Failure Conditions are assessed on wet runway conditions.

$$\text{failure\_conditions\_assessed\_wet}$$

SASP 1.1-3 RTO is considered an operational condition caused either by an external event or by a system failure (annunciated or perceived by the flight crew) during takeoff run.

$$\text{rto\_caused\_external\_or\_failure}$$

SASP 1.1-4 The flight crew will not initiate an RTO due to an annunciated failure of the deceleration function after V1.

$$(\text{annunciated\_failure\_decel\_after\_v1} \Rightarrow \neg\text{rto\_initiated})$$

SASP 1.1-5 Taxi is performed at groundspeeds below 30 knots.

$$\text{groundspeed\_taxi\_kt} < 30$$

SASP 1.1-6 Overrunning the runway length at or above "XYZ" knots is considered a high speed overrun.

$$(\text{groundspeed\_overrun\_kt} \geq \text{xyz\_overrun\_threshold\_kt} \Rightarrow \text{high\_speed\_overrun})$$

SASP 1.1-7 Overrunning the runway length below "XYZ" knots is considered a low speed overrun.

$$(\text{groundspeed\_overrun\_kt} < \text{xyz\_overrun\_threshold\_kt} \Rightarrow \text{low\_speed\_overrun})$$

SASP 1.1-8 Failures of deceleration capability will be detected and annunciated by on-board systems.

$$(\text{failure\_decel\_capability} \Rightarrow \text{detected\_and\_annunciated\_onboard})$$

2 Axioms

$$\text{flight\_duration\_hours} = 5$$

$$\text{p\_loss\_elec\_bus\_per\_flight} = \text{lambda\_loss\_elec\_bus\_per\_hour} \cdot \text{flight\_duration\_hours}$$

$$\text{p\_loss\_pedal\_per\_flight} = \text{lambda\_loss\_pedal\_per\_hour} \cdot \text{flight\_duration\_hours}$$

3 Internal consistency

- `PSSA_REQUIREMENTS`: SAT
- `ALT_PSSA_REQUIREMENTS`: SAT
- `SPEC_REQUIREMENTS`: SAT
- `BSCU_PSSA_REQUIREMENTS`: SAT
- `BSCU_ASSUMPTIONS_TO_WBS`: SAT
- `BSCU_SPEC_REQUIREMENTS`: SAT
- `AIRPLANE_REQUIREMENTS`: SAT
- `PASA_SAFETY_REQUIREMENTS`: SAT
- `AIRPLANE_ASSUMPTIONS`: SAT
- `WBS_SFHA_ASSUMPTIONS`: SAT

4 Directional entailment

`AIRPLANE_REQUIREMENTS` entails each `PASA_SAFETY_REQUIREMENTS`

- **`PASA-SR-01`**: PASS
- **`PASA-SR-05`**: PASS
- **`PASA-SR-10`**: PASS
- **`PASA-SR-12`**: PASS
- **`PASA-SR-13`**: PASS
- **`PASA-SR-14`**: PASS

Summary: 6 passed, 0 failed.

`SPEC_REQUIREMENTS` entails each `PSSA_REQUIREMENTS`

- **`E12-1`**: PASS
- **`E12-2`**: PASS
- **`E12-3`**: PASS
- **`E13-1`**: PASS
- **`E13-2`**: PASS
- **`E13-3`**: PASS
- **`E13-4`**: PASS
- **`E13-5`**: PASS
- **`E13-6`**: PASS

- `E13-7`: FAIL
  text: When "HYD 1 Enable" output is disabled, then "Alt/Emer Ctrl" output shall be enabled
  witness: `alt_emer_ctrl_on=False, hyd1_enable_on=False`
- `E13-8`: PASS
- `E13-9`: PASS
- `E14-1`: PASS
- `E14-2`: PASS
- `E14-3`: FAIL
  text: The probability of 7 or more wheel speed sensors erroneous or inoperative will be less than 1.0E-07 per flight
  witness: `p_seven_or_more_sensors_per_flight=1.0000e-07`
- `E14-4`: FAIL
  text: The probability of loss of an airplane electrical power bus will be less than 1.0E-04 per flight
  witness: `p_loss_elec_bus_per_flight=1.0000e-04`
- `E14-5`: FAIL
  text: The probability of loss of a Left brake pedal position input will be less than 1.0E-04 per flight
  witness: `p_loss_pedal_per_flight=1.0000e-04`
- `E14-6`: PASS
- `E14-7`: PASS

Summary: 15 passed, 4 failed.

`BSCU_SPEC_REQUIREMENTS` entails each `BSCU_PSSA_REQUIREMENTS`

- `BSCU-001`: PASS
- `BSCU-002`: PASS
- `BSCU-003`: PASS
- `BSCU-004`: PASS
- `BSCU-005`: PASS
- `BSCU-006`: PASS
- `BSCU-007`: PASS
- `BSCU-008`: PASS
- `BSCU-009`: PASS
- `BSCU-010`: PASS
- `BSCU-011`: PASS

Summary: 11 passed, 0 failed.

5 Restatement equivalence

- **`E14-1` ⇔ `E28-WBS-ASMP-1`**: EQUIVALENT
- **`E14-2` ⇔ `E28-WBS-ASMP-2`**: EQUIVALENT
- **`E14-3` ⇔ `E28-WBS-ASMP-3`**: EQUIVALENT
- **`E14-4` ⇔ `E28-WBS-ASMP-4`**: NOT EQUIVALENT
  `E14-4`: The probability of loss of an airplane electrical power bus will be less than 1.0E-04 per flight
  `E28-WBS-ASMP-4`: The failure rate for loss of an airplane electrical power bus will be less than 1.0E-04 per hour of flight
  `E14-4` entails `E28-WBS-ASMP-4`: PASS
  `E28-WBS-ASMP-4` entails `E14-4`: FAIL; witness: `p_loss_elec_bus_per_flight=4.8828e-04`
- **`E14-5` ⇔ `E28-WBS-ASMP-5`**: NOT EQUIVALENT
  `E14-5`: The probability of loss of a Left brake pedal position input will be less than 1.0E-04 per flight
  `E28-WBS-ASMP-5`: The failure rate for loss of a left brake pedal position input will be less than 1.0E-04 per hour of flight
  `E14-5` entails `E28-WBS-ASMP-5`: PASS
  `E28-WBS-ASMP-5` entails `E14-5`: FAIL; witness: `p_loss_pedal_per_flight=4.8828e-04`
- **`E14-6` ⇔ `E28-WBS-ASMP-6`**: EQUIVALENT
- **`E14-7` ⇔ `E28-WBS-ASMP-7`**: EQUIVALENT
- **`ASMP 3.2.2-6` ⇔ `SASP 1.1-5`**: EQUIVALENT
- **`ASMP 3.2.2-8` ⇔ `SASP 1.1-8`**: EQUIVALENT
- **`ASMP 3.2.2-1` ⇔ `SASP 1.1-6`**: NOT EQUIVALENT
  `ASMP 3.2.2-1`: Overrunning the runway length above "XYZ" knots is considered a high speed overrun.
  `SASP 1.1-6`: Overrunning the runway length at or above "XYZ" knots is considered a high speed overrun.
  `ASMP 3.2.2-1` entails `SASP 1.1-6`: FAIL; witness: `high_speed_overrun=False, xyz_overrun_threshold_`
  `groundspeed_overrun_kt=0`
  `SASP 1.1-6` entails `ASMP 3.2.2-1`: PASS

Summary: 7 equivalent, 3 not equivalent.